

Project Carrier Course, part 1
Lecture 6: Project Introduction

CM2015 - Autumn 2025 - Martin Jacobsson

Three Flavours

- **Biomechanics**

  17 students 4 groups
- **Imaging**

  14 students 4 groups
- **Health Systems**

  3 students 1 group

Schedule and Deadlines

Project start: October 27 (or soon afterwards)
Seminar 1: November 12 (compulsory attendance for everyone)

- 8:15-9:00: Biomechanics groups
- 9:15-10:00: The rest

Seminar 2: December 11 (compulsory attendance for everyone)

- 8:15-9:00: The rest
- 9:15-10:00: Biomechanics groups

Deadline 1 report: December 18 noon
Deadline peer review feedback: January 2
Poster deadline: January 7 (for printing)
Presentation: January 9 (compulsory attendance for everyone)
Deadline final report: January 13
Deadline individual assignment: January 12

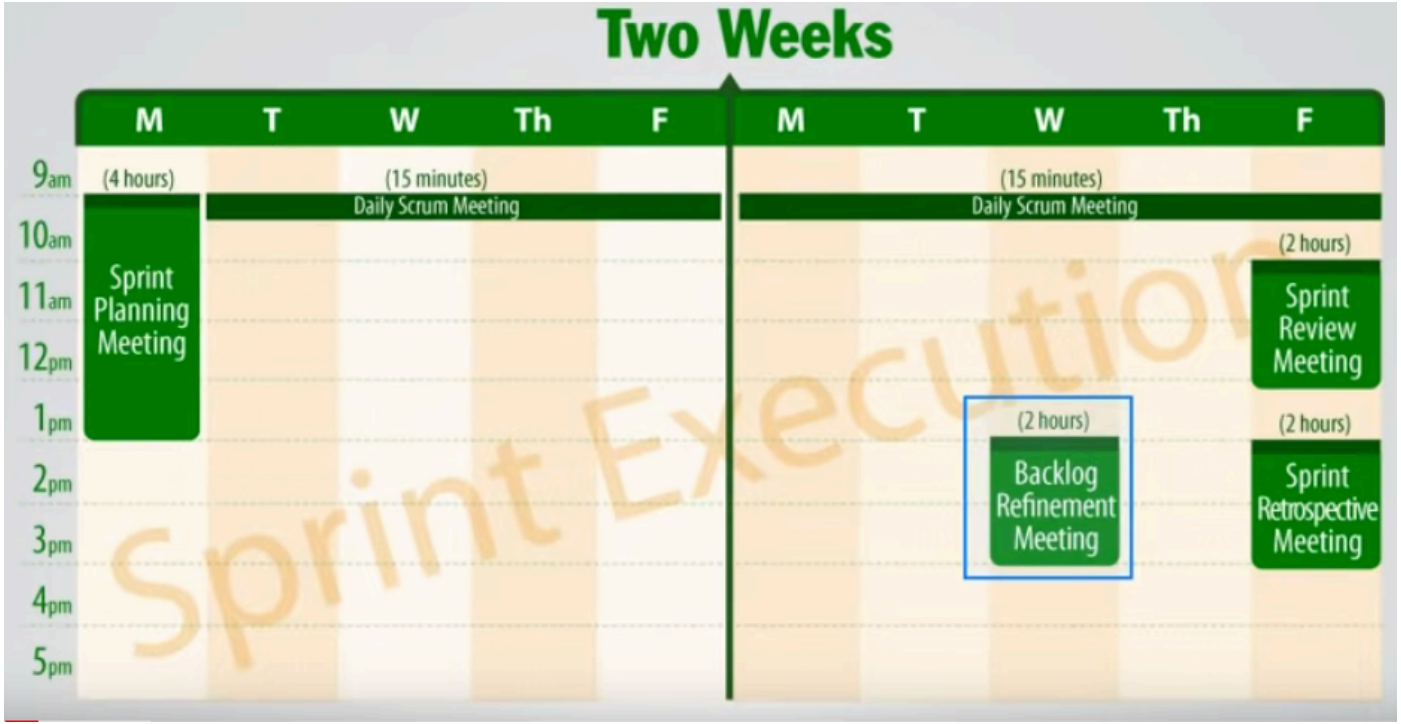
Ethics Seminar

Submit ethical section on November 7.

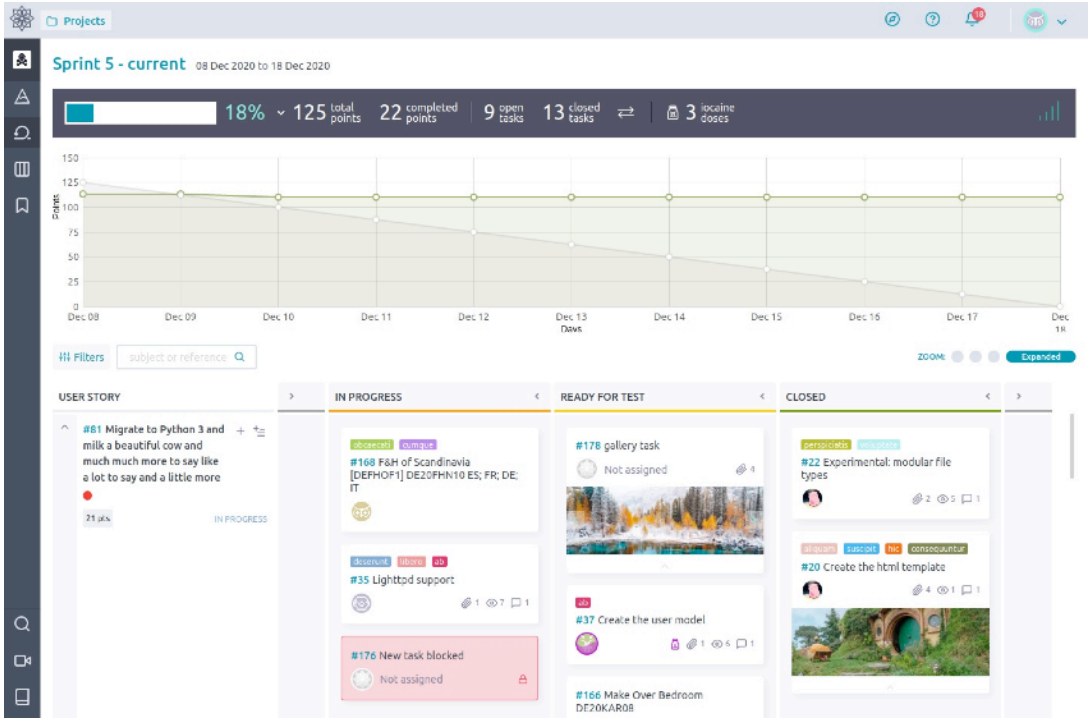
We distribute the texts to everyone in the course.

Seminar: Tuesday 11 November 11:15.
Active discussions about the ethical sections at the seminar.

Scrum - Sprint Overview



Scrum Board (post-it notes on a wall)



Creating a User Story



User stories have a specific format:

"As a **[persona]**,
I want to **[do something]**
so that I can **[realize a reward]**"

Who is the user?

Why do they want to do this?
What is the benefit/reward?

https://www.alexandercowan.com/best-agile-user-story/

The Projects

BM-Child — Segmentation of growing neck bones in infants towards impact injury analysis

Extends previous work on infant cervical facet joint analysis to a broader paediatric population

Medical image segmentation and statistical morphological analysis of CT images

Siyuan Chen, Xiaogai



BM-Vibration - An app for measuring vibration exposure time

Mikael Forsman, Michail Magkos



Hand-arm vibration injury syndrome (HAVS)

Using a wearable, measure: exposure and type of activity





BM-Dentist

Maksim Kornevs

Bluetooth stress ball to signal pain at the dentist







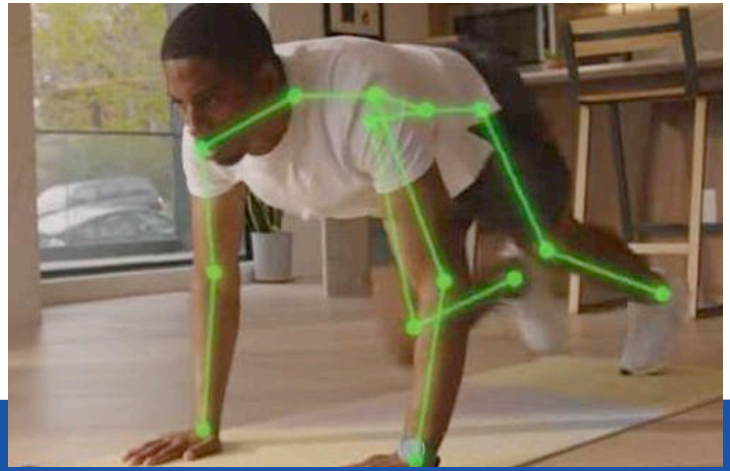
BM-IMUVideo



Jonas Willén

Sync skeleton tracking with IMU/Force plates

For use in rehabilitation, physiotherapy, ergonomics, and sports





IMG-Phantom - Dynamic Mathematical Phantoms for PET Image Reconstruction

Mamo Colarieti-Tosti

Synthetic Phantoms for Positron Emission Tomography (PET)

Develop a computational framework for generating dynamic phantoms:

- Simulate physiologically inspired time-varying activity distributions based on compartmental models.
- Provide flexible control over kinetic parameters and geometric structures.
- Serve as an open-source tool for evaluating reconstruction algorithms and generating synthetic training datasets for deep learning-based methods.

IMG-Dosimeter

Kian Shaker, Mamo

Measure occupational X-ray exposure

Develop and validate an energy-integrating measurement mode in the next OpenDosimeter firmware (v.1.1) that will extend the device's operational range to medium and high dose rates.



IMG-SWE - Advanced Ultrasound Technique for Muscle Characterization

Estelle Pitti, Lorena Claeys

Shear Wave Elastography (SWE)

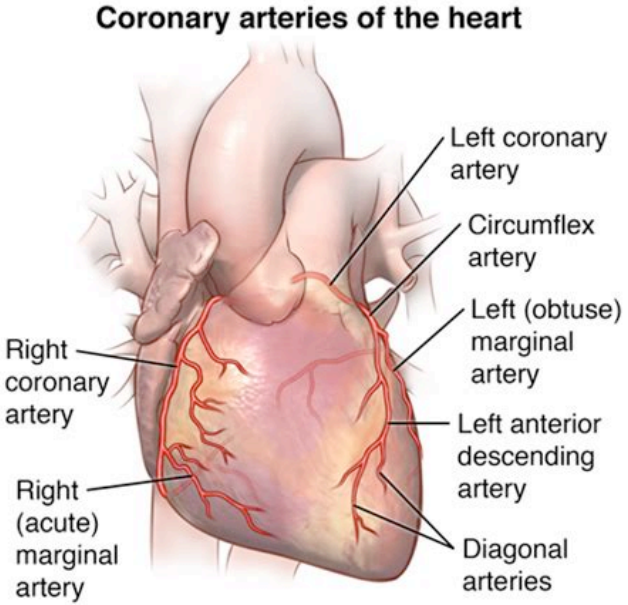
While SWE works well for isotropic tissues like the liver, skeletal muscle is transversely isotropic (TI), meaning its stiffness differs along and across the fibers.

The project involves acquiring data for an ongoing SWE study.

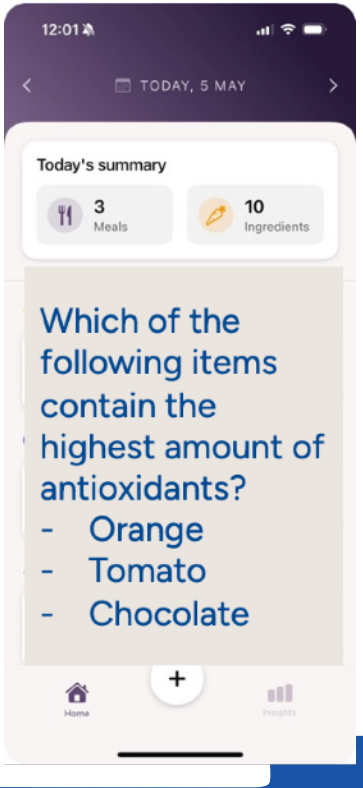
IMG-Heart - Image-based geometric descriptors of the coronary artery tree

Eva Breznik

Given CT scans and corresponding segmented masks develop and implement a number of local vessel anatomy descriptors.



Health Systems



Martin Jacobsson

pQBL - purse Question-Based Learning

For instructing people with dietary problems





How to do the project?!

What to do?

A literature study

Do something for real, build a prototype

Write a report

Peer review

Poster presentation

Organize the work according to Scrum

Most Effort Here!

The Report

See the Course Memo: https://www.kth.se/kurs-pm/CM2015/20252/50755

The project must also deliver a written report that describes the project's aim, design, and results. In addition, relevant **ethical and sustainability aspects** must be included. There must be one section highlighting each student's contribution to the report and project. The report should be written in the same style as a **final thesis** report. Hence, it is advised to use such a template and follow the same structure, unless you have good reasons not to. **There is no page limit** on the report, but it needs to contain the relevant parts and should avoid unnecessary repetition. Note that usually **a long report is not better**. A report should only contain relevant parts, avoid unnecessary repetition, and describe everything in a compact and concise fashion. Be sure to **present a research question** and everything in the report should address that question. Hence, no need to explain a lot of half-relevant stuff.

Also grading criteria…

Who is the Reader?

- Expectations from the reader
  - Conformity with other articles, why?
- What does the reader likely already know?
  - How much lacking background knowledge does the reader need/want to look up?
- What does the reader want to gain from the article?
  - Why does the reader read the article?
- You read a many articles?
  - What did you expect? What background knowledge did you look up? Why did you read them?

Peer-review

Every group gets two other reports to review.

Need for constructive feedback.

Use this template:

- Title of the paper
- What are the strengths of the paper? What are the contributions?
- How is the presentation of the paper? Well-structured, good language, easy to follow, and at right level for the intended audience?
- Does the paper make a good use of references? Too few? Too many? The right ones?
- Does the paper make a good use of figures, tables, and formulas? Too few? Too many? The right ones?
- How does the paper deal with its chosen topic/problem/question? Do you agree with its approach and conclusions? How can it be improved?
- Other comments to the authors

How to Find Relevant Material

Literature Search

State-of-the-Art Search

State-of-the-Art Search

Wikipedia (or LLMs)

- You get a good overview from here.
- So start here!
- **But**: Do not reference the wikipedia article. It makes your article look like a “school project”!
- **Instead**: Follow the references outlined in the Wikipedia article. Maybe reference them?

Types of (Scientific) Articles

Conference, Workshop Paper
- Short, Focused, Often work in progress

Journal Article
- Longer, More Complete Work, Can be very advanced

Magazine Article
- Less details, More Popular Science, Shorter

White Paper
- Popular Science, Introduction to an Area
- Written for Practitioners

News Articles

Standards
- Long, Extremely detailed, No explanations

Research Article Search Services

KTH Library www.kth.se/biblioteket

Google Scholar, ACM Digital Library, PubMed, ResearchGate

DiVA, Libris

Typical services:

- Search
- References (used in the article)
- Citations (which other articles refer to this one)
- List by author

Scientific Social Networks and Search













AI-based Tools















And many many more …

keep popping up as mushrooms in the forest…

Search Strategies

- Search first and skim through, read later

- Keyword-based search
- Find a land-mark paper (or Wikipedia article), follow the references, both reference list and citations lists
- Look at relevant conferences/journals and look at their technical program or table of contents.
- Find relevant researchers or research groups.

- Collect the articles and the citation in the process.

Good Research, Bad Research

- Not all research papers are good.
- There is plenty of crap!!
- How to tell them apart:
  - Famous conference, journal, ...
  - High Field Impact Factor (FIF)
  - Famous co-author
  - Lot’s of citations
  - Analyse the paper



How to Make a Poster

Poster Content

Keep in mind the topic and the objective of the task:

- *present to you fellow students a specific topic from the specific perspectives of PCC1*

Focusing on the questions of the task.

Include that information in a poster layout

- not exceeding 500 words of text
- use figures (but reference them when you “borrow” them)

This is a Catchy Title

Eat THIS or DIE!

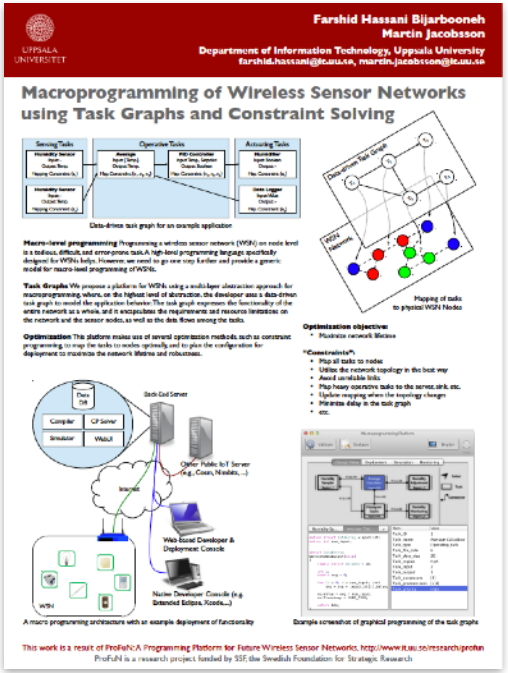
The poster has two proposes:

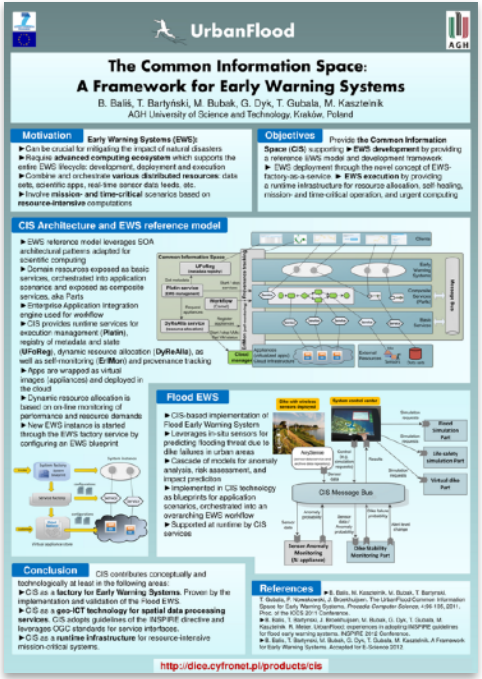
1. Catch the attention in an overcrowded place
2. Convey the key message in an efficient way

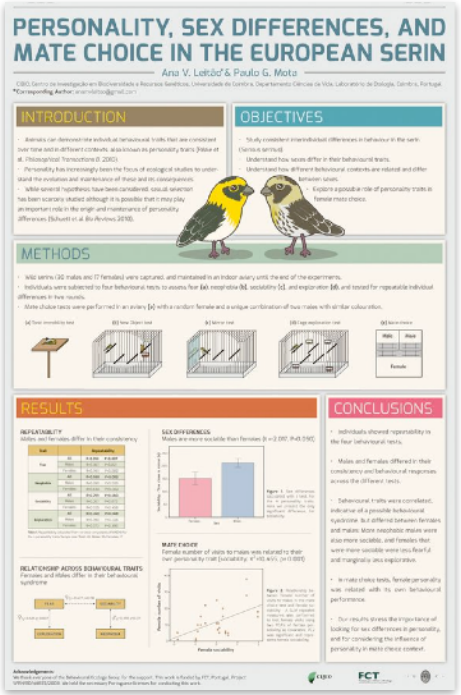
- Use a catchy title and **good figures**. Not too much text
- Invite to interaction with the poster presenter
- Use the poster when you explain key concepts

The 7 Habits of Very Healthy People

Poster Examples







Vancouver Reference Style

Book

[1] Davino C, Furno M. Quantile regression: theory and applications. Wiley; 2013.

Journal

[2] Brown CD, Davis HT. Receiver operating characteristics curves and related decision measures: a tutorial. Chemometrics and Intelligent Laboratory Systems. 2006; 80(1): 24-38.

Conference

[3] Kabashima Y, Obuchi T, Uemura M. Approximate cross-validation formula for Bayesian linear regression. In: 2016 54th Annual Allerton Conference on Communication, Control, and Computing (Allerton). Monticello (IL), USA. 27-29 Sept. 2016, p. 596-600.

Online

[4] Ulrich J. TTR: Technical trading rules. https://cran.rproject.org/web/packages/TTR/index.html (accessed 23 April 2017).

Further Reading

Short Video Tutorial

https://www.youtube.com/watch?v=PLuMRqcZhrI

Online Resource

https://connect.le.ac.uk/posters

More Videos

https://www.youtube.com/watch?v=k6FzrwW3oKQ
https://www.youtube.com/watch?v=GJwcVWszLC4
https://www.youtube.com/watch?v=A4qXONix2aQ

Printing

Print format: A1 (594 × 841 mm)

Upload PDF (with all fonts included, “Embedded Subset”)
Minimum font size: 12pt
Avoid dark color backgrounds
Deadline: January 7
We print it, we bring pins
You pin it up on a poster screen

No KTH logo on the poster!
(not allowed by KTH)

Poster Presentation and Demo

Date: Friday 9 January, 10:00 - 12:00 (time is preliminary)
Place: Probably in the atrium

- We take turns to present and listen/interact
- The whole session is compulsory for all students